

Simulation and Synthetic Data Generation of Deformable Objects

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Motivation

- Getting large high-quality datasets is overall difficult
- This is especially the case if we want to work with transparent objects and even more if we want them deformed
- There are papers that already solve this issue for generation of non-transparent objects and also for glass objects (this approach is easily applicable to any transparent object)

Goal

- To create an application or several scripts that allow for automated creation of rendered images of deformed transparent bottles (e.g. plastic bottle) for synthetic dataset creation

Related Work

- Automated 3D Dataset Generation for Arbitrary Objects – a work that proposes a workflow for aligning scanned meshes onto a background to create synthetic data
- Synthetic Dataset Rendering with Parametric Transparent and Translucent Objects – a bachelors thesis that proposes a workflow for creating synthetic data for transparent objects and also how to set parameters for realistic lighting
- Clear Grasp – method for position estimation of translucent objects
- ISAAC Sim – environment for creation and rendering of various types of synthetic data

Methodology (depending on approach)

- Blender 5.0
- Unreal Engine 5 – still a possibility in final steps
- All scripts will be written in python 3.11 or 3.12 depending on the need for compatibility with bpy

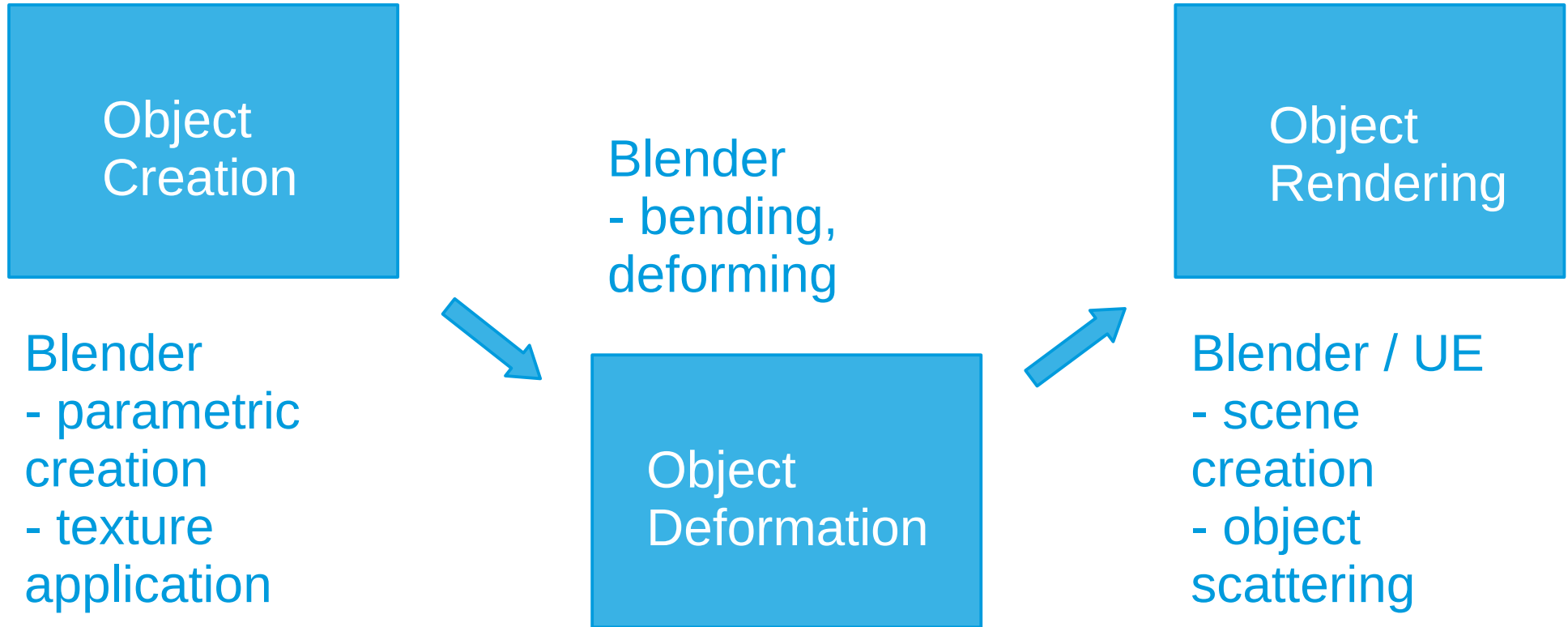
Blender

- free and open-source 3D computer graphics software tool
- used for creating animated films, visual effects, art, 3D-printed models, motion graphics, interactive 3D applications, and virtual reality

Unreal Engine

- a 3D computer graphics game engine
- Unreal Engine 5 features Nanite, a virtualized geometry system that allows game developers to use arbitrarily high quality meshes with automatically generated Level of Detail, and Lumen, a dynamic global illumination and reflections system that uses software and hardware ray tracing

Workflow



Images



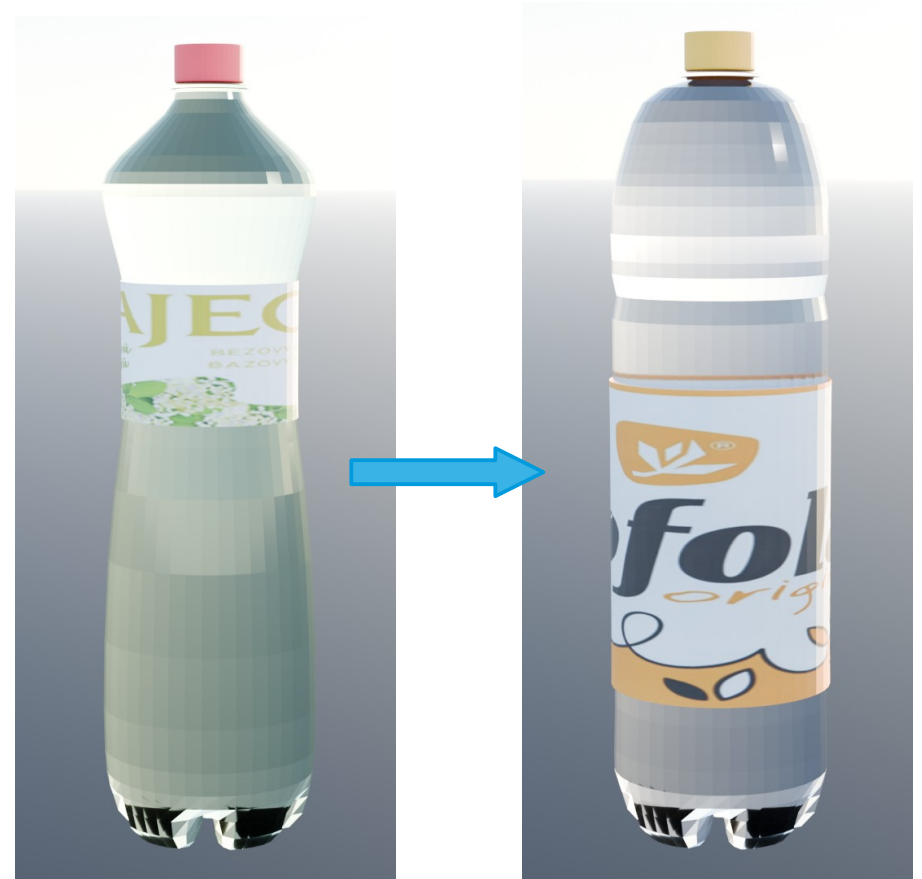
Example scene from
Automated 3D Dataset
Generation for Arbitrary
Objects



Example scene from
Synthetic Dataset
Rendering with Parametric
Transparent and
Translucent Objects

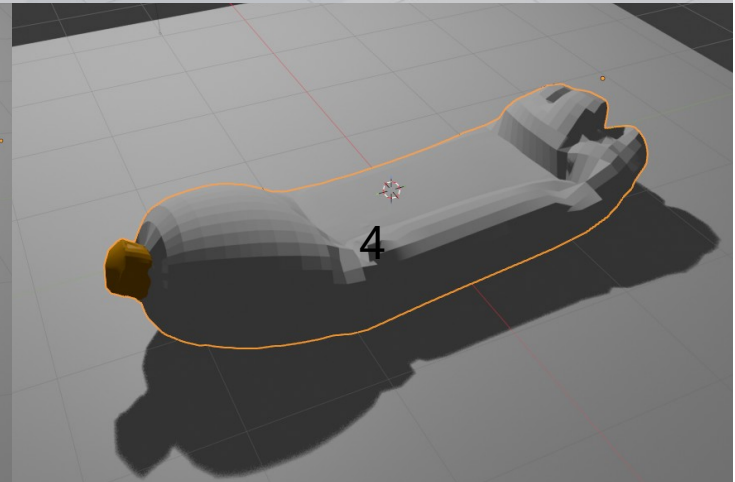
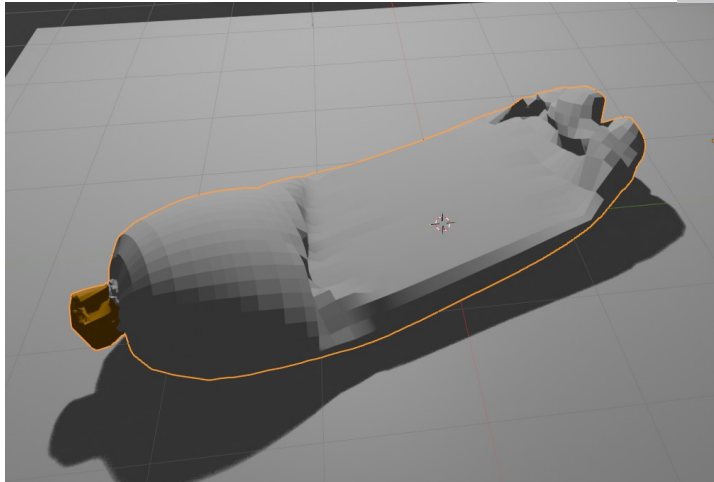
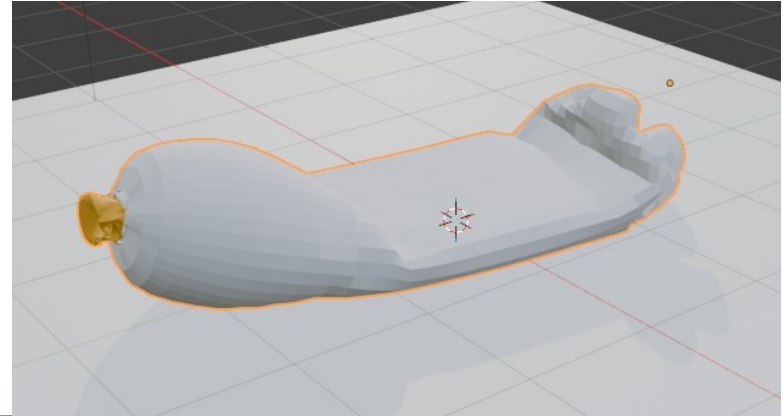
Object Creation

- 29 ring parameters
- bottle cap color
- label texture
- plastic color



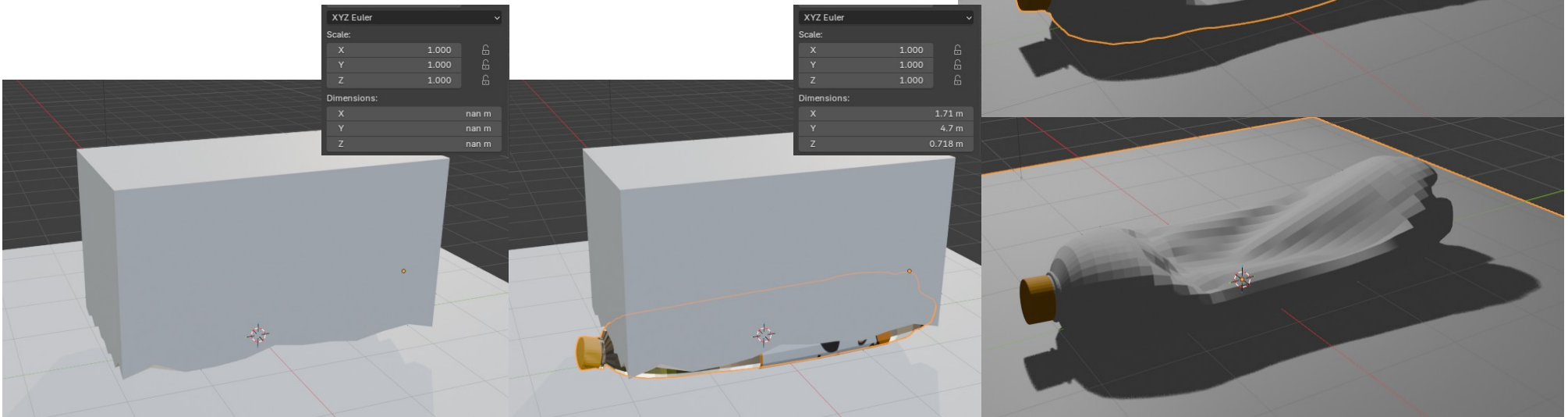
Object Deformation: Cloth Physics

- cloth object with internal pressure
- not very good, has numerous issues



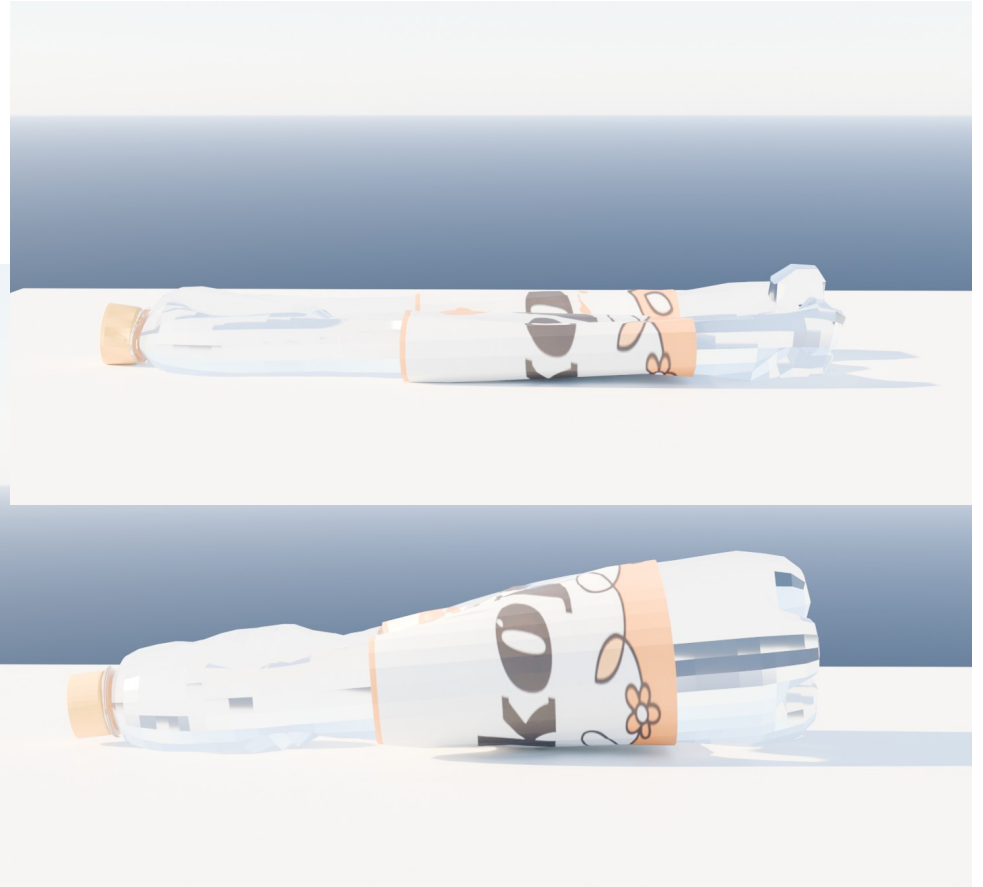
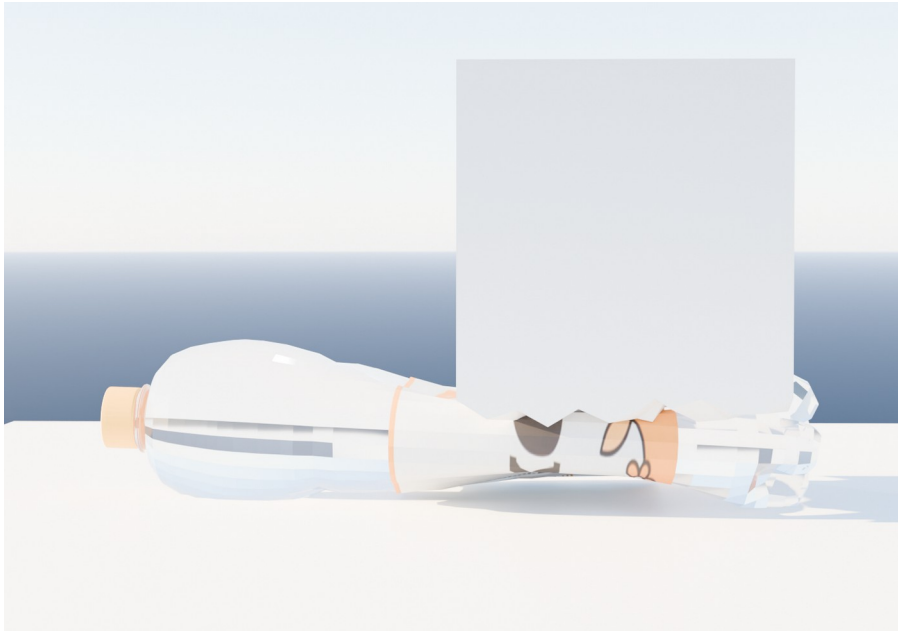
Object Deformation: Soft-body Physics

- our go-to
- realistic but has some issues



Object Rendering

- currently placeholder rendering using Cycles



Future Objectives

- explore various rendering approaches, Blender shaders, Mitsuba, ...
- object scattering using the likes of mentioned related works
- evaluate the accuracy of our created data compared to reality
- (potentially) generate and test a dataset

Thank you for your attention

References

- Synthetic Dataset Rendering with Parametric Transparent and Translucent Objects - Michal Kubirita, Comenius University in Bratislava Faculty of Mathematics, Physics and Informatics, 2024
- Automated 3D Dataset Generation for Arbitrary Objects - Paul Schulz , Thorsten Hempel, Ayoub Al-Hamadi, 2017
- Surface Reconstruction of Transparent Objects from Gaussian Radiance Fields - Peter Kravar, 2025